

Literature Status for arXiv:2205.13287

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Status

Result type: literature_already_answered.

Verdict: a weak-star counterpart question left open in arXiv:2205.13287 has a negative answer in later literature.

Original Question

The source paper is:

Rainis Haller, Eve Oja, and Mart Poldvere, *Diameter two properties for spaces of Lipschitz functions*, arXiv:2205.13287.

After separating several non-weak-star diameter two properties for spaces of Lipschitz functions, the paper records that it remained open whether these properties coincide with their weak-star versions. In particular, it asks whether, for spaces of Lipschitz functions, w^* -SSD2P implies slice-D2P.

Supporting Answer

The supporting paper is:

Rainis Haller, Jaan Kristjan Kaasik, and Andre Ostrak, *Separating diameter two properties from their weak-star counterparts in spaces of Lipschitz functions*, arXiv:2404.11430.

The supporting paper explicitly identifies the unresolved weak-star counterpart issue from the Haller–Oja–Poldvere paper. In Section 3 it gives a metric space

$$M = \{a_1, a_2, b_k, c_k : k \in \mathbb{N}\}$$

with distances $d(a_1, b_{2k-1}) = d(a_2, b_{2k}) = d(c_k, b_l) = 1$ for $k \leq l$, and distance 2 between different points in all remaining cases.

For this metric space, the authors verify that M has SLTP. By the known characterization, this yields w^* -SSD2P for $\text{Lip}_0(M)$. They then define a functional

$$G = \frac{1}{2}(m_{a_1, a_2} + F), \quad F(f) = \lim_{\mathcal{U}} f(m_{b_{2n-1}, b_{2n}}),$$

where $\mathcal{U}$ is a free ultrafilter, and use the slice $S(G, \alpha)$ to show that the diameter of this slice is strictly less than 2 for suitable α . Thus $\text{Lip}_0(M)$ does not have the slice-D2P.

Consequently w^* -SSD2P does not imply slice-D2P for spaces of Lipschitz functions. This negatively answers the selected open question from arXiv:2205.13287.

Scope And Limitations

This is an already-known literature answer, not a new proof produced in this run. It closes the weak-star-counterpart question selected from arXiv:2205.13287, and it separates w^* -SSD2P from slice-D2P within the class of spaces $\text{Lip}_0(M)$.

It does not settle the separate question in arXiv:2205.13287 asking whether there exists a metric space M such that $\text{Lip}_0(M)$ has slice-D2P but not D2P. The later paper arXiv:2404.11430 still lists the distinction between (w^*) -slice-D2P and (w^*) -D2P in Lipschitz function spaces as open.

Run Evidence

The lightweight run indexes were searched for arXiv ids 2205.13287 and 2404.11430 and for the phrases weak-star counterparts, w^* -SSD2P, and slice-D2P. A prior packet using arXiv:2404.11430 exists for the distinct 2018 dual-space question about w^* -SSD2P without SSD2P, but no packet for this exact arXiv:2205.13287 source question was found.

References

- [1] R. Haller, E. Oja, and M. Poldvere, *Diameter two properties for spaces of Lipschitz functions*, arXiv:2205.13287.
- [2] R. Haller, J. K. Kaasik, and A. Ostrak, *Separating diameter two properties from their weak-star counterparts in spaces of Lipschitz functions*, arXiv:2404.11430.